



NOSQL DATABASE DESIGN – LAB SHEET - 3

1. Design a Student Activity Tracking System in MongoDB that stores student details, enrolled courses, attendance, skills, and recent activities. Perform queries to retrieve students based on different activity conditions.
2. Develop an Event Logging System that records login, logout, failed-login, file-upload, and password-change events. Use MongoDB queries to identify repeated failed-login attempts and frequently occurring events.
3. Create a Mini Blogging Platform using MongoDB where each blog contains author, category, tags, comments, and publication information. Retrieve blogs using different combinations of tags and categories.
4. Design a Flexible Product Catalog where laptops, mobiles, and cameras have different sets of attributes. Store them in a common collection and write queries to search category-specific properties.
5. Develop an E-Commerce Order Management System containing customers, products, and orders. Write queries to retrieve a customer's complete purchase history and calculate order totals.
6. Create a Website Analytics System that stores page visits, user ID, device, browser, location, and timestamp. Use aggregation to identify the most visited pages and commonly used devices.
7. Design a Content Management System for storing articles, authors, categories, tags, and publication status. Write queries to retrieve published and unpublished content according to different criteria.
8. Implement a Shopping Cart System using embedded MongoDB documents. Perform operations for adding products, changing quantities, removing products, and calculating the cart value.
9. Create a Product Review and Rating System in MongoDB. Store customer reviews and use aggregation to calculate the average rating and number of reviews for each product.
10. Design an Inventory Monitoring System containing product stock, minimum stock level, supplier, and restocking date. Write queries to identify products that need immediate restocking.



11. Implement a Bank Wallet Transfer System using MongoDB transactions. Transfer an amount between two users and ensure that both balance updates are completed atomically.
12. Create a User Notification System for messages, order updates, and system alerts. Write queries to retrieve unread notifications and update them after the user views them.
13. Design a Course Learning Platform containing courses, instructors, students, modules, progress, and completion status. Retrieve students according to their course progress.
14. Create a Real-Time Order Tracking System where an order moves through stages such as Placed, Confirmed, Shipped, and Delivered. Store the complete status history and retrieve the current status of an order.
15. Develop a Flexible News Database in which different news articles may contain different fields such as author, location, category, tags, images, and sources. Write queries to search articles despite variations in document structure.

